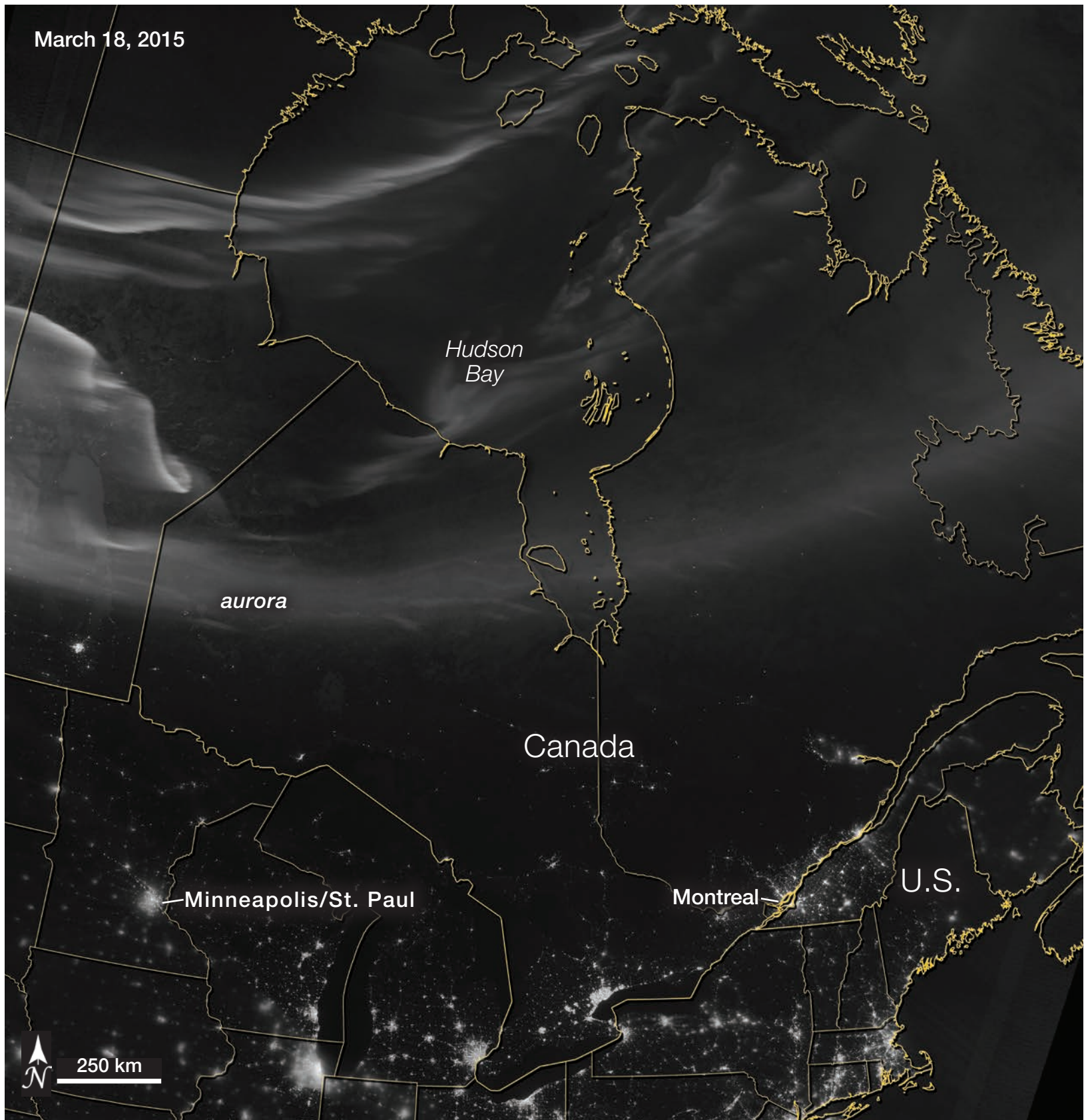
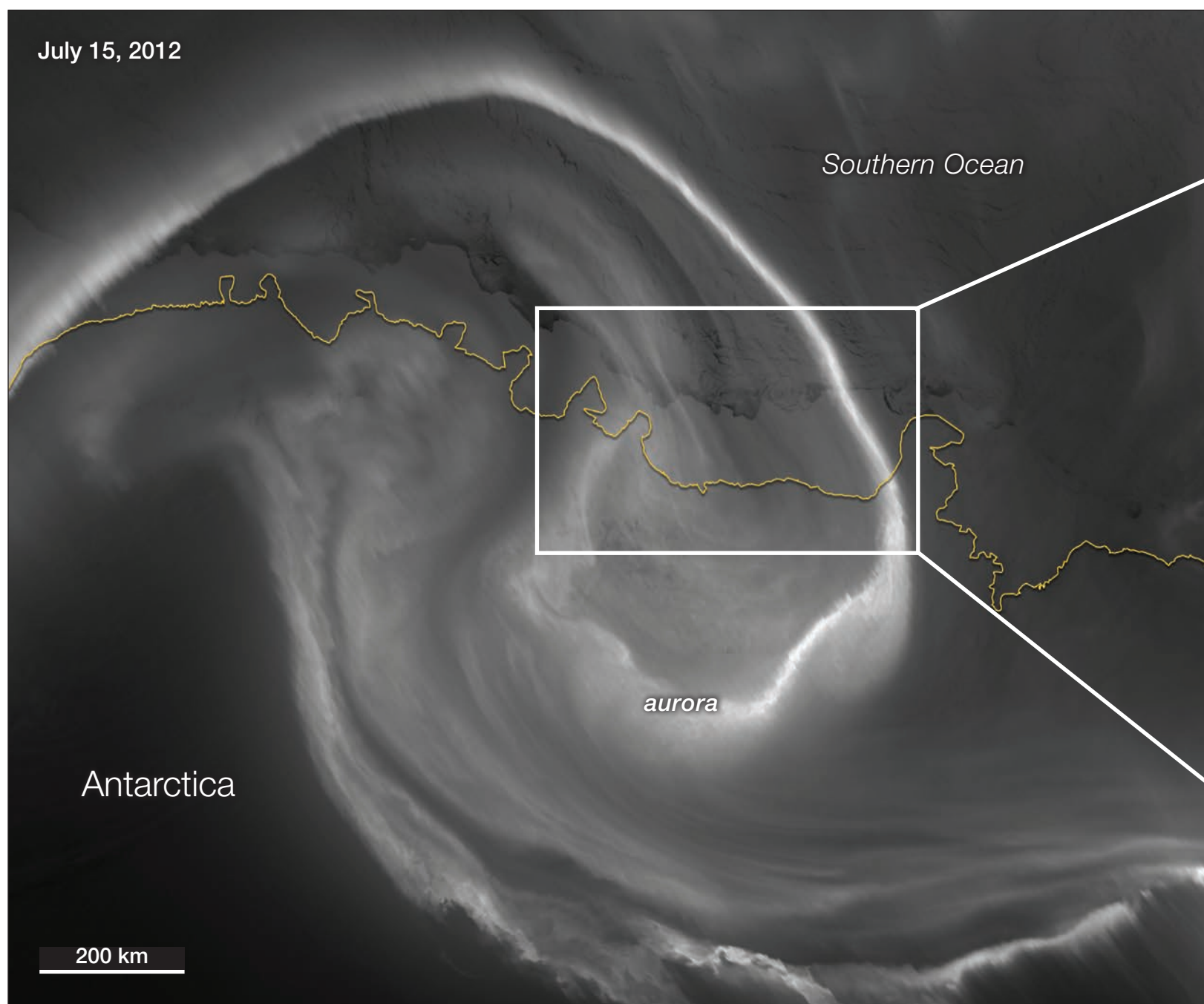




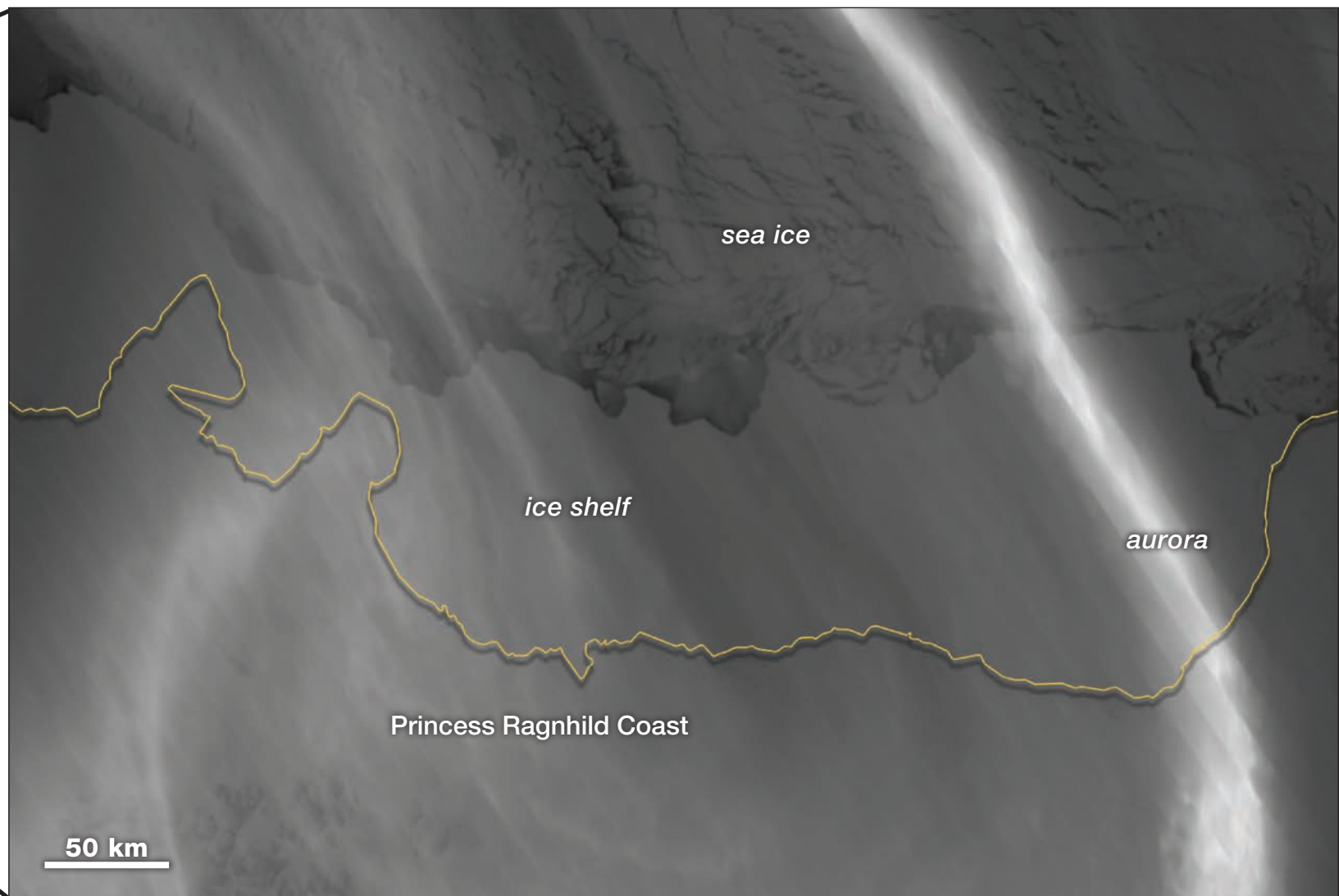


Aurora Lights Up the Night—Antarctica

On July 15, 2012, the VIIRS DNB on the Suomi NPP satellite captured this night-time view of the aurora australis over Antarctica's Queen Maud Land and the Princess Ragnhild Coast. The slightly jagged appearance of the auroral lines is a function of the rapid dance of the energetic particles at the same time that the satellite is moving and



the VIIRS sensor is scanning. The white box in the left image depicts the area shown in the close-up image at right (below). Light from the aurora was bright enough to illuminate the ice edge between the ice shelf and the Southern Ocean. At the time, Antarctica was locked in midwinter darkness and the Moon was a waning crescent that provided little light.





Airglow

Airglow—Australia

On October 7, 2018, an astronaut onboard the ISS shot this photograph at an altitude of about 250 miles (400 kilometers) over Australia. In this view, stars appear more numerous along the image center where the plane of the disk-shaped Milky Way galaxy extends into space. The orange colors enveloping Earth are known as airglow—diffuse bands of light that stretch roughly 50 to 400 miles (80 to 645 kilometers) into our atmosphere. This type of airglow is known as chemiluminescence, or nightglow. More than just a pretty light show, airglow reveals some of the workings of the upper reaches of our atmosphere. Studying airglow can help scientists learn about the movement of particles near the interface between Earth and space, including the connections between space weather and Earth weather. Airglow reveals some of the conditions of the upper atmosphere, such as its temperature, shape, and the amounts of different types of gases.

October 7, 2018





April 2, 2014





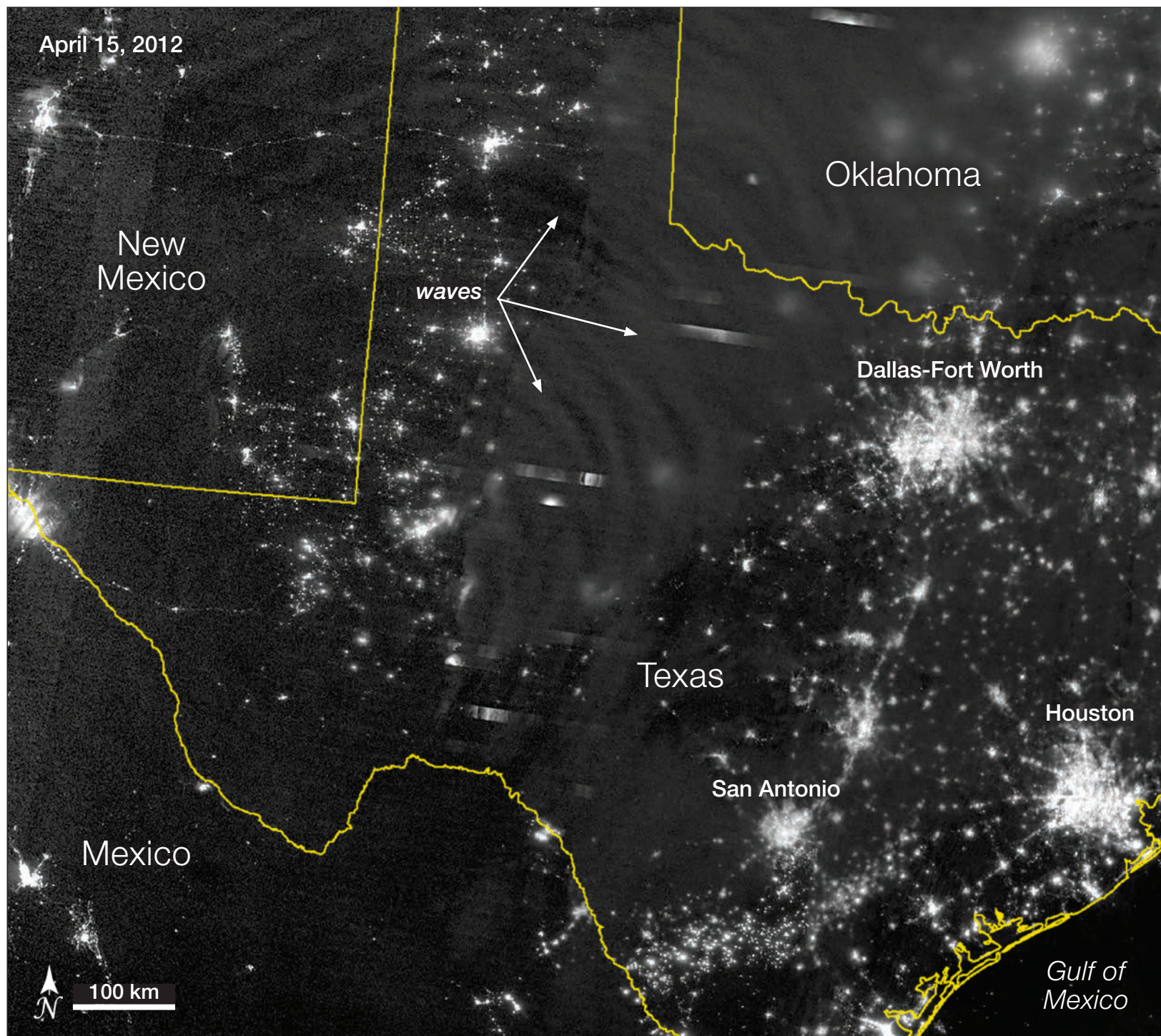
Night Colors—Russia

A thin green line of the aurora borealis crosses the top of this photograph, taken by an astronaut on the ISS on April 2, 2014. The Moon appears as a white disc just above the aurora. Airglow appears as a blue-white cusp along Earth's limb. Russia's capital city, Moscow, makes a splash of yellow light and is easily recognized by the radial pattern of its highways. Other cities are Nizhny Novgorod [250 miles (400 kilometers) from Moscow], Saint Petersburg [388 miles (625 kilometers) away], and Finland's capital city Helsinki [about 545 miles (876 kilometers) away].

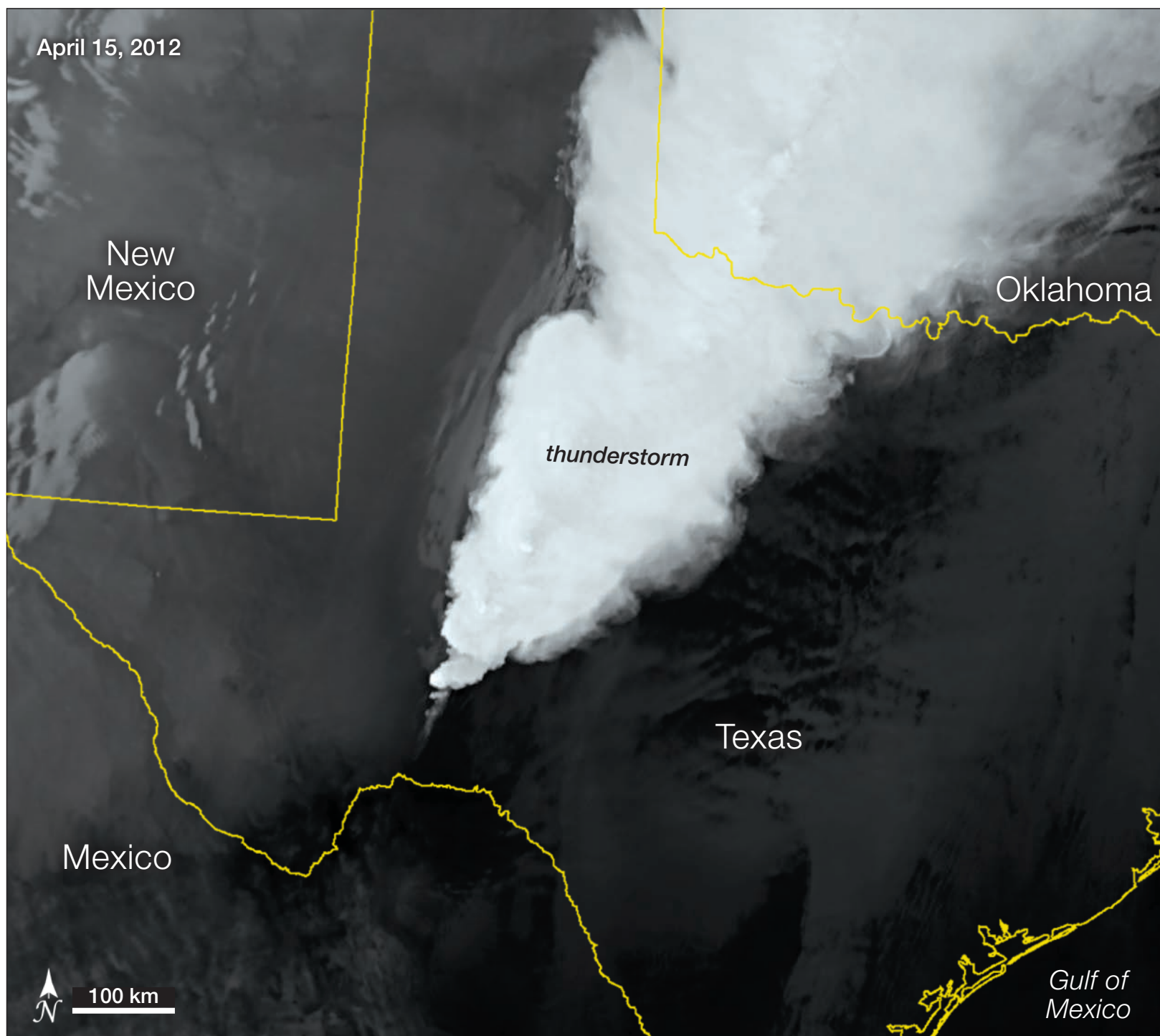


Waves in Airglow—Texas

As shown in the left image (below), in April 2012 waves in Earth's airglow spread across the nighttime skies of Texas like ripples in a pond. In this case, the waves were provoked by a massive thunderstorm. The VIIRS DNB on the Suomi NPP satellite captured these glowing ripples in the night sky on April 15, 2012. The right image (next page)



shows the thunderstorm as observed by a VIIRS thermal infrared band. This thermal band, which is sensitive only to heat emission, is not sensitive to the subtle visible-light wave structures seen by the VIIRS DNB. Clouds appear white because they are cooler than Earth's surface.





[Opposite page] This photograph, taken by an astronaut from the ISS on December 10, 2017, shows the city of Bangkok, the capital and largest city in Thailand, illuminated by city lights. The adjacent waters of the Andaman Sea and Gulf of Thailand are illuminated by hundreds of green lights on fishing boats. Fishermen use the lights to attract plankton and fish, the preferred diet of commercially important squid. As the bait swims to the surface, the squid follow to feed and get caught by fishermen. The same fishing practices are used off the Atlantic coast of South America. The border between Thailand and neighboring Cambodia to the east is distinguished by a marked difference in the number of city lights. Cambodia is less urbanized and its population is smaller than that of neighboring countries. The majority of Cambodia's population lives in rural farming areas where electricity is sparse.

Human Light Sources

Viewing Earth at night from space reveals a distinctly human story. As one gazes upon the points of light in images captured from orbiting sensors, discerning eyes begin to see the fingerprints of human presence: from the glow of megacities and the pinpoints of bright light produced by fleets of fishing boats at sea, to gas flares glowing like candles in the darkness, lights show where we have made our homes, established industries, mined natural resources, and built roads. Patterns are everywhere.

It quickly becomes evident looking at nightlight data that human activity is highly influenced by Earth's natural features and processes, or physical geography. For example, the vital link between humans and water is quite clear. Nearly half the world's population lives along or near coastlines. Others live along rivers and freshwater bodies. Most population centers tend to be along coastlines, near rivers, or near major over-land transportation hubs (e.g., highways, railroads) that cut slivers of light through the darkness between cities. In general, our species tends to avoid settling in regions with harsh climates (or if we do, we set about transforming them into something more pleasant), in mountainous areas with steep terrain, and where basic resources like food and water are limited (e.g., deserts). Accordingly, more light is clustered around regions where the climate is moderate, terrain is flat, resources are plentiful, and soil is fertile.

Areas with the brightest light are generally the most urbanized—but not necessarily the most populated. When a city or country is thriving, electricity is used to keep businesses, schools, and factories bustling with activity. Lights show how far urban sprawl extends, as well as areas where growth is occurring or where growth has not yet occurred. Cities usually have many people concentrated in a small area, so electricity usage is high. Poor areas may have large populations but low availability or use of electric lights. These signals from anthropogenic nightlights provide clear indicators of how we've settled our planet and—with some analysis—what we do with our energy.